

ERC Dynamic Accessibility Lattice (DAL): Fixed Scaffold, Dynamic Traversability under Perturbation–Recovery Constraints

Exploratory computational note

This document accompanies a reproducible computational notebook and presents an exploratory, falsification-oriented operational framework for studying dynamic accessibility under perturbation–recovery constraints.

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Abstract

This exploratory computational note examines whether systems may preserve structural scaffold while progressively reorganizing which futures remain operationally accessible under perturbation–recovery constraints.

Using a minimal lattice-based framework with fixed structural connectivity and dynamically changing traversability, we explore whether perturbation timing, recovery kinetics, perturbation history, and stochastic variability may reshape effective accessibility without requiring explicit structural disconnection.

Accessibility is evaluated through operational observables including reachable future fraction, accessibility cost, and weighted efficiency under repeated perturbation regimes. We further examine whether organization across parameter space remains observable under robustness tests, null controls, and stochastic perturbations.

Across the explored parameter ranges, accessibility organization appears operationally consistent with interactions between perturbation timing and recovery dynamics while structural scaffold remains preserved. Importantly, this work is intended as a minimal computational abstraction for exploratory and falsification-oriented analysis rather than a claim regarding universal physical or biological mechanisms.

The framework is computationally reproducible and accompanies an executable notebook, exported figures, and structured data tables.

1. Introduction

Many systems appear capable of preserving large-scale structural organization while progressively changing how coordination, recoverability, or effective responses remain operationally accessible under repeated perturbation.

In practice, functional deterioration does not necessarily coincide with abrupt structural loss. Instead, systems may preserve scaffold while progressively reorganizing which coordinated futures remain reachable, recoverable, or dynamically accessible.

This distinction motivates a minimal operational question:

Can accessibility reorganize while scaffold remains preserved?

Rather than assuming that accessibility degradation necessarily requires explicit structural disconnection, this work explores whether perturbation timing, recovery kinetics, and

perturbation history may be sufficient to progressively reorganize operational accessibility within a fixed scaffold.

Importantly, the objective is intentionally exploratory and falsification-oriented. The framework is not introduced as a claim regarding universal physical or biological mechanisms, but rather as a minimal computational abstraction designed to test whether accessibility organization may emerge operationally under perturbation–recovery constraints.

2. Motivation and operational question

The present notebook operationalizes a minimal question:

Can operational accessibility reorganize while structural scaffold remains preserved?

Rather than assuming that accessibility degradation necessarily requires explicit structural disconnection, the framework explores whether perturbation timing, recovery kinetics, perturbation history, and stochastic variability may progressively reorganize operational accessibility within a fixed scaffold.

To investigate this question, the notebook evaluates accessibility through a minimal set of operational observables:

- reachable future fraction,
- accessibility cost,
- weighted efficiency.

These observables are examined under repeated perturbation regimes, timing variation, recovery heterogeneity, parameter sweeps, and falsification-oriented controls.

The objective is intentionally exploratory rather than explanatory.

Instead of proposing a universal mechanism, the framework asks whether accessibility organization may emerge operationally in a minimal computational system under perturbation–recovery constraints while scaffold remains structurally preserved.

Throughout the notebook, emphasis is placed on observable organization, robustness testing, null controls, stochastic perturbation, and computational reproducibility.

3. Minimal framework: fixed scaffold, dynamic traversability

To explore whether accessibility may reorganize without requiring structural removal, we begin from a minimal computational abstraction consisting of a fixed scaffold and dynamically changing traversability.

The scaffold is represented as a persistent lattice-like interaction structure whose connectivity remains unchanged throughout the simulation. Under this minimal setup, structural organization is intentionally preserved.

What changes operationally is not scaffold existence itself, but the effective traversability of interactions. Under repeated perturbation, local connections may become progressively more or less traversable depending on system state, thereby reshaping which futures remain operationally reachable without explicit structural disconnection.

The central distinction is intentionally minimal: **preserved scaffold does not necessarily imply preserved accessibility.**

Accessibility is evaluated operationally through reachable future fraction, accessibility cost, weighted efficiency, and effective traversability under perturbation.

At this stage, the objective is intentionally descriptive rather than explanatory:

to test whether accessibility organization may emerge under perturbation–recovery constraints while scaffold remains structurally preserved.

Figure 1 — Fixed scaffold with dynamic effective accessibility

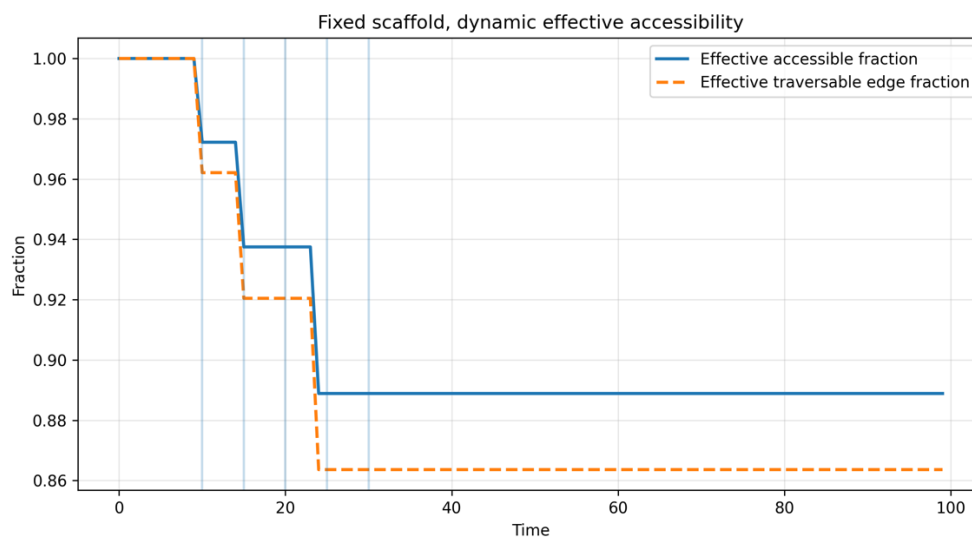


Figure 1. The figure illustrates the distinction between preserved scaffold and changing effective accessibility. While structural organization remains fixed, operational accessibility evolves dynamically under repeated perturbation

4. Reachable futures from a fixed source

Having established a fixed scaffold with dynamically changing traversability, the next operational question becomes:

Which futures remain reachable from a fixed origin as traversability evolves?

Rather than treating accessibility as a binary property of structural connectivity, the framework evaluates accessibility from the perspective of a fixed source embedded within a preserved scaffold.

Operationally, accessibility is quantified through the fraction of nodes that remain reachable from a fixed origin under changing traversability conditions.

Under repeated perturbation, systems may preserve structural scaffold while progressively reorganizing which regions remain operationally reachable from a given state.

The central operational distinction becomes: **preserved connectivity does not necessarily imply preserved reachable futures.**

The objective at this stage is descriptive rather than explanatory: to examine whether repeated perturbation progressively reorganizes accessibility while scaffold remains structurally preserved.

Figure 2 — Reachable futures under changing traversability

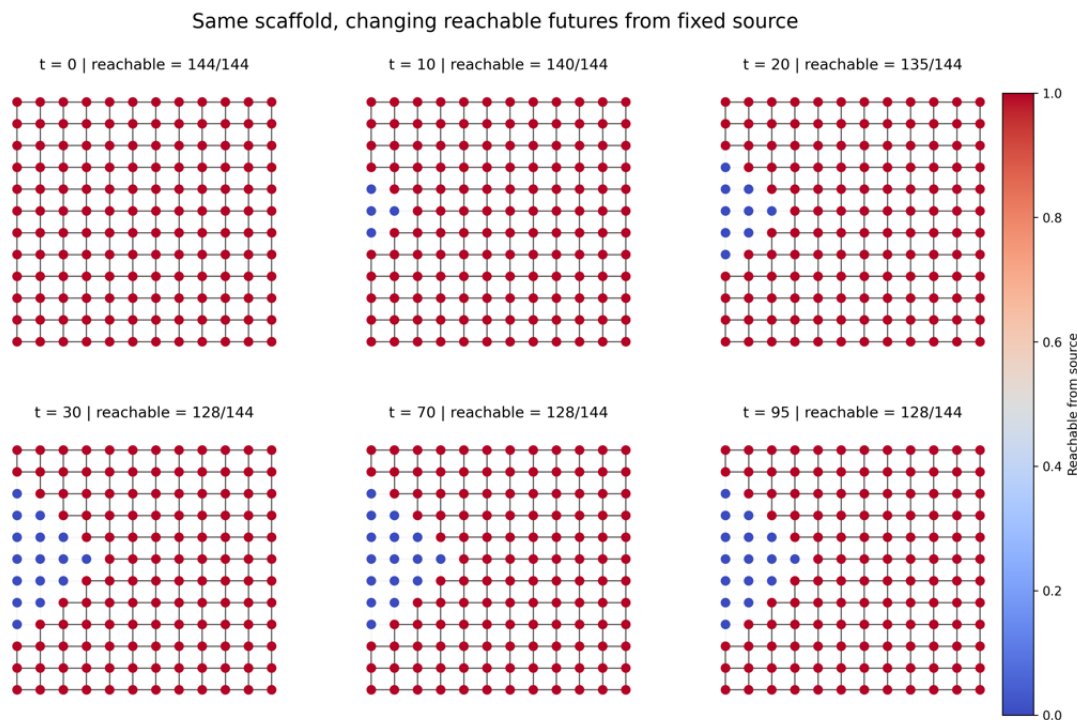


Figure 2. The snapshots illustrate how reachable regions may progressively reorganize from a fixed source despite preservation of the underlying scaffold. Structural organization remains present while operational accessibility evolves dynamically through time.

5. Timing-dependent accessibility

Having established that reachable futures may reorganize under preserved scaffold, the next operational question becomes:

Does perturbation timing reshape accessibility under equal perturbation magnitude?

To explore this question, the notebook compares perturbation schedules that preserve comparable total perturbation burden while varying temporal delivery structure.

Under this setup, perturbations are redistributed across time while maintaining similar cumulative exposure, allowing accessibility to be examined as a function of timing rather than total perturbation alone.

Across explored schedules, accessibility appears sensitive to perturbation timing. Clustered and temporally distributed perturbations may reorganize reachable futures differently despite comparable overall perturbation burden.

The operational distinction becomes: **identical perturbation burden does not necessarily imply identical accessibility futures**

The objective at this stage remains descriptive rather than explanatory: to examine whether perturbation timing alone may reorganize accessibility while scaffold remains preserved.

Figure 3 — Timing-dependent accessibility under equal perturbation burden

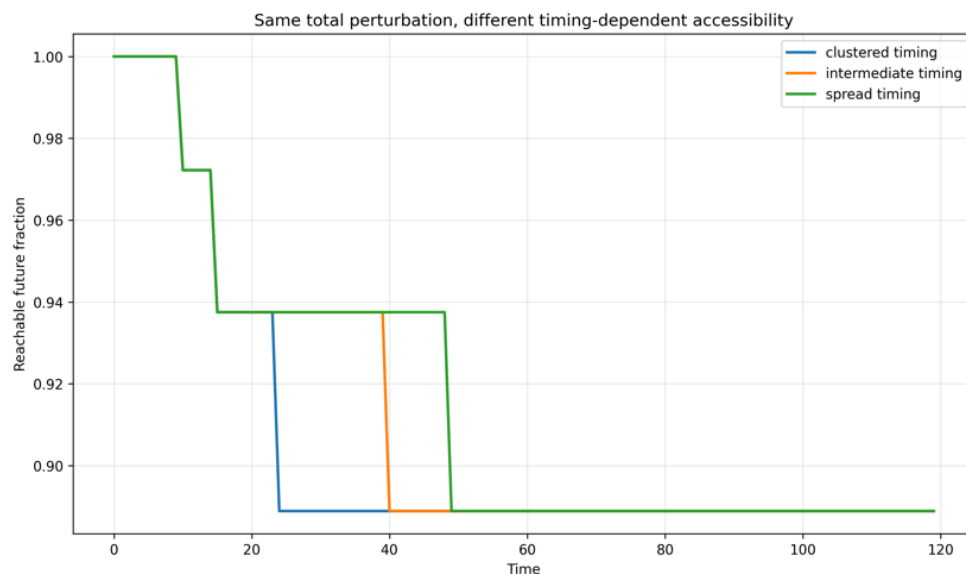


Figure 3. The figure illustrates how accessibility evolves under perturbation schedules with comparable total burden but distinct temporal organization. Differences emerge operationally through accessibility trajectories while scaffold remains preserved.

6. Accessibility cost under repeated perturbation

The previous sections examined whether perturbation timing may reorganize reachable futures within a preserved scaffold.

A related operational question then becomes:

How costly does accessibility become while scaffold remains preserved?

Rather than treating accessibility as purely binary (reachable versus unreachable), the framework evaluates whether coordination progressively becomes more operationally costly under repeated perturbation.

To explore this distinction, accessibility is examined through complementary observables including mean accessibility cost, worst-case accessibility cost, and weighted accessibility efficiency.

Under repeated perturbation, traversability may progressively constrain how efficiently accessible regions remain coordinated within a structurally preserved scaffold.

Across explored schedules, accessibility cost frequently increases while weighted efficiency declines, even when scaffold remains structurally present.

The operational distinction becomes: **preserved scaffold does not necessarily imply preserved coordination efficiency.**

At this stage, the objective remains descriptive rather than explanatory:

to examine whether accessibility degradation may emerge progressively through increasing coordination burden rather than explicit structural removal.

Figure 4 — Accessibility cost and weighted efficiency under repeated perturbation

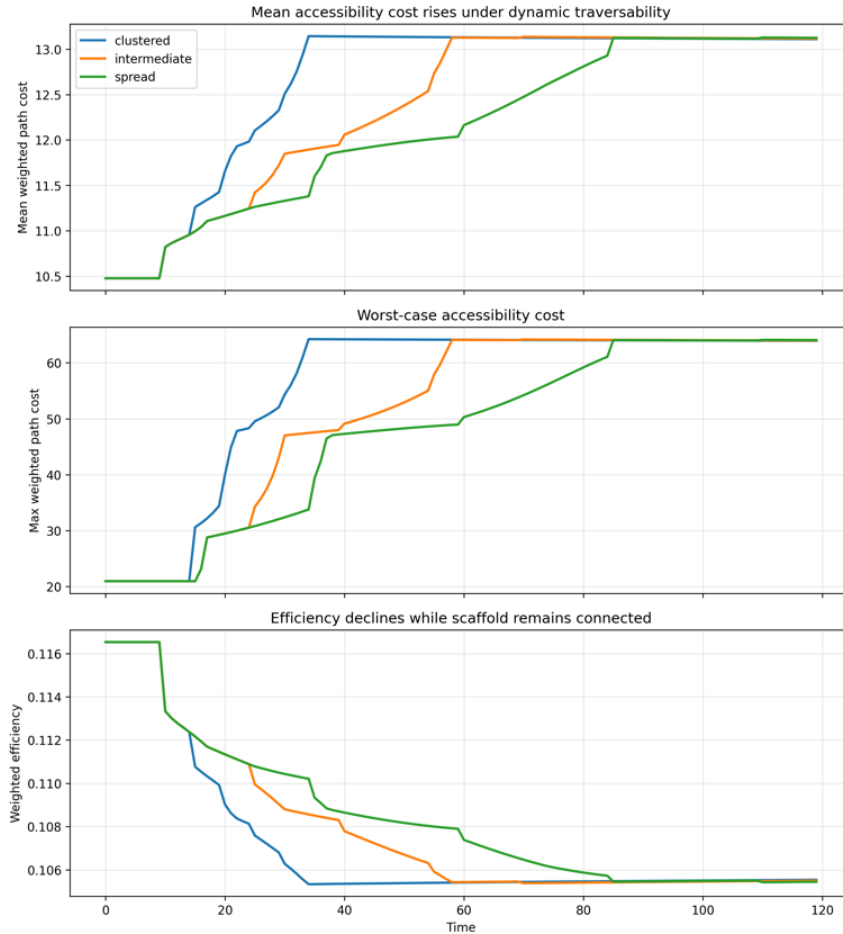


Figure 4. The figure illustrates how accessibility cost and efficiency evolve while scaffold remains structurally preserved. Under explored perturbation schedules, accessibility may progressively reorganize through increasing coordination burden and declining weighted efficiency rather than explicit structural disconnection.

7. Recovery kinetics reshape accessibility

The previous sections explored how perturbation timing may reorganize accessibility and how coordination burden may progressively increase under repeated perturbation.

A further operational question then becomes:

Do identical perturbations reorganize accessibility differently under distinct recovery kinetics?

To explore this possibility, the notebook compares systems exposed to identical perturbation schedules while varying recovery dynamics.

Under this setup, scaffold structure, perturbation timing, and perturbation burden remain fixed while recovery-related parameters are allowed to differ operationally.

Across explored conditions, accessibility appears sensitive to recovery kinetics. Faster and slower recovery conditions may reorganize reachable futures, accessibility cost, and weighted efficiency differently despite otherwise comparable perturbation schedules.

The operational distinction becomes: **identical perturbations do not necessarily imply identical accessibility futures when recovery kinetics differ.**

At this stage, the objective remains descriptive rather than explanatory: to examine whether recovery kinetics alone may reorganize effective accessibility under otherwise comparable perturbation conditions.

Figure 5 — Accessibility organization under distinct recovery kinetics

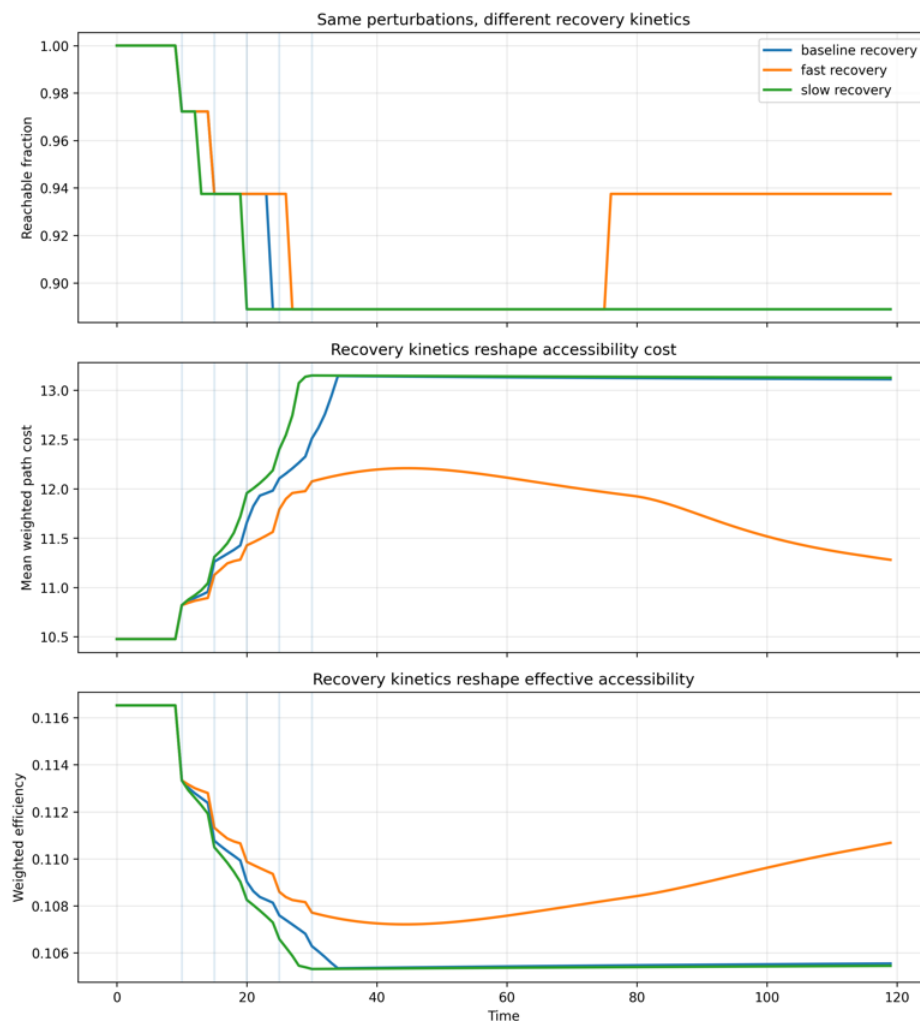


Figure 5. The figure illustrates how systems exposed to identical perturbation schedules may display distinct accessibility trajectories under different recovery conditions. Differences emerge operationally through reachable future fraction, accessibility cost, and weighted efficiency while scaffold remains preserved.

8. Emergent organization in parameter space

The previous sections examined individual perturbation schedules and recovery conditions.

A broader operational question then becomes:

Does accessibility organization emerge systematically across perturbation and recovery conditions?

To explore this possibility, the notebook evaluates accessibility across a parameter space spanning perturbation interval and recovery kinetics.

Rather than focusing on isolated examples, the objective is to examine whether operational organization becomes observable across multiple perturbation–recovery combinations.

The emphasis remains intentionally operational: **the objective is not to infer physical phases or universal transitions, but to examine whether interpretable accessibility organization emerges in parameter space.**

Three complementary observables are evaluated:

- weighted accessibility efficiency,
- accessibility cost,
- reachable future fraction.

Across explored conditions, accessibility organization appears structured with respect to perturbation timing and recovery dynamics.

Regions associated with slower recovery and denser perturbation schedules frequently coincide with reduced weighted efficiency, increased accessibility cost, and contraction of reachable futures.

Conversely, faster recovery and lower perturbation density frequently remain associated with comparatively preserved accessibility organization.

The operational distinction becomes: **accessibility organization appears dependent on the interaction between perturbation timing and recovery dynamics.**

8.1 Weighted accessibility efficiency

Weighted efficiency summarizes how effectively accessibility remains operationally organized under evolving traversability.

Figure 6 — Weighted accessibility efficiency across parameter space

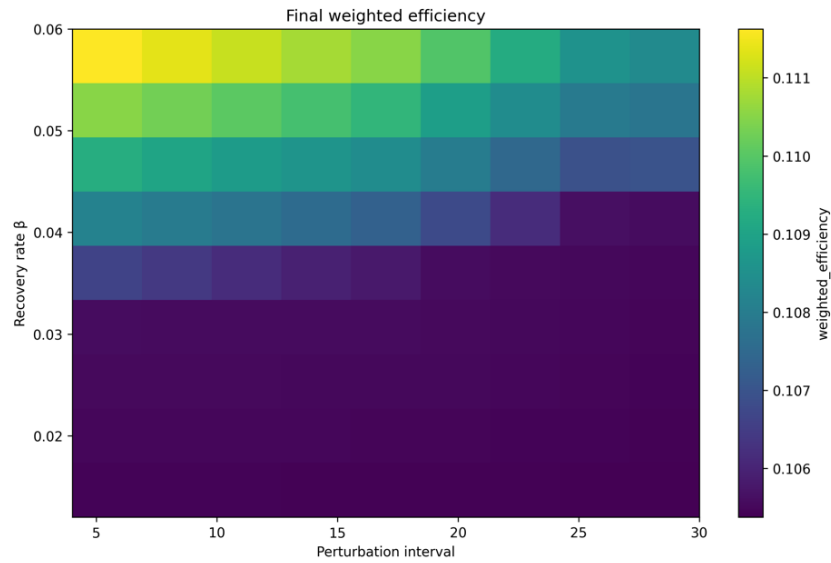


Figure 6. Weighted efficiency provides an operational summary of how effectively accessibility remains organized under perturbation–recovery constraints. Variation appears gradual across explored conditions rather than through abrupt binary transitions.

8.2 Accessibility cost

Accessibility cost quantifies how coordination burden changes as traversability evolves.

Figure 7 — Accessibility cost across parameter space

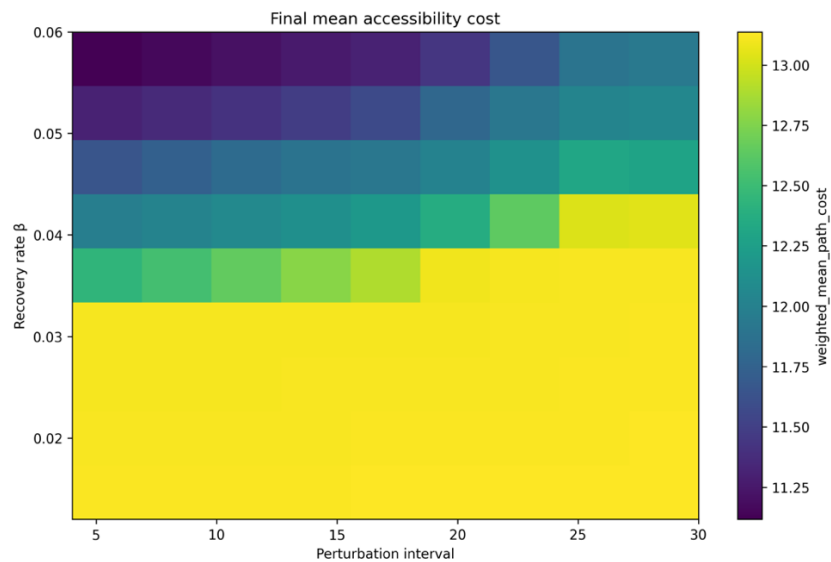


Figure 7. Accessibility cost provides a complementary operational perspective by quantifying how coordination burden changes under repeated perturbation. Regions of reduced accessibility frequently coincide with increased operational cost.

8.3 Reachable future fraction

Reachable future fraction summarizes the proportion of scaffold that remains operationally accessible from a fixed source.

Figure 8 — Reachable future fraction across parameter space

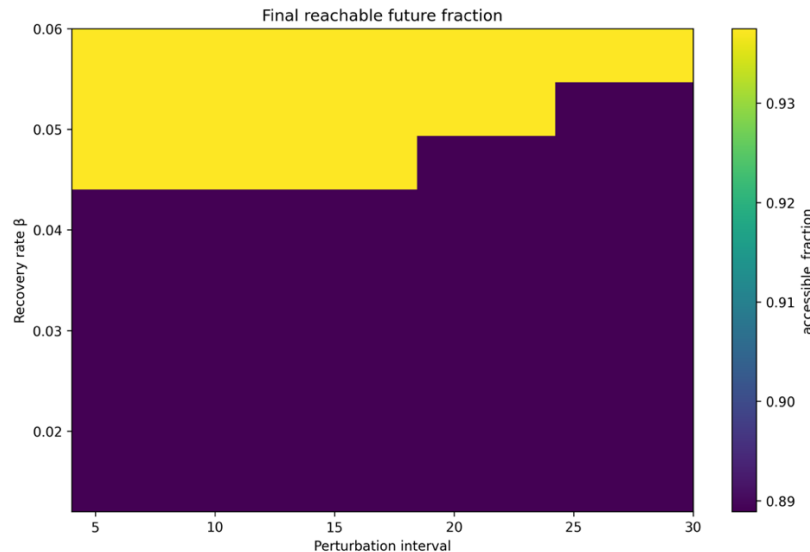


Figure 8. Reachable future fraction provides an interpretable estimate of accessibility contraction under perturbation–recovery constraints. Although more threshold-sensitive than weighted observables, it summarizes contraction of reachable operational futures.

9. Robustness and falsification-oriented controls

The previous sections suggest that accessibility organization may emerge under perturbation–recovery constraints.

However, observational organization alone is insufficient.

A central objective of this notebook is therefore falsification-oriented: **to test whether observed accessibility organization survives attempts to weaken, simplify, or perturb underlying assumptions.**

Rather than treating observed organization as inherently meaningful, the notebook evaluates whether organization remains observable under:

- threshold variation,
- simplified recovery dynamics,
- reduced perturbation-history persistence,
- and stochastic perturbation variability.

The operational question becomes:

Does accessibility organization remain observable under perturbed assumptions?

Importantly, the objective is not to prove organization, but rather: **to determine which aspects of organization remain operationally admissible under progressively weakened conditions.**

9.1 Threshold robustness

Operational question

Does accessibility organization depend on arbitrary threshold choice?

To examine whether organization emerges only under a specific accessibility threshold, the notebook evaluates organization across multiple threshold choices.

The objective is intentionally conservative: **to test whether qualitative organization survives threshold variation.**

Figure 9 — Threshold robustness

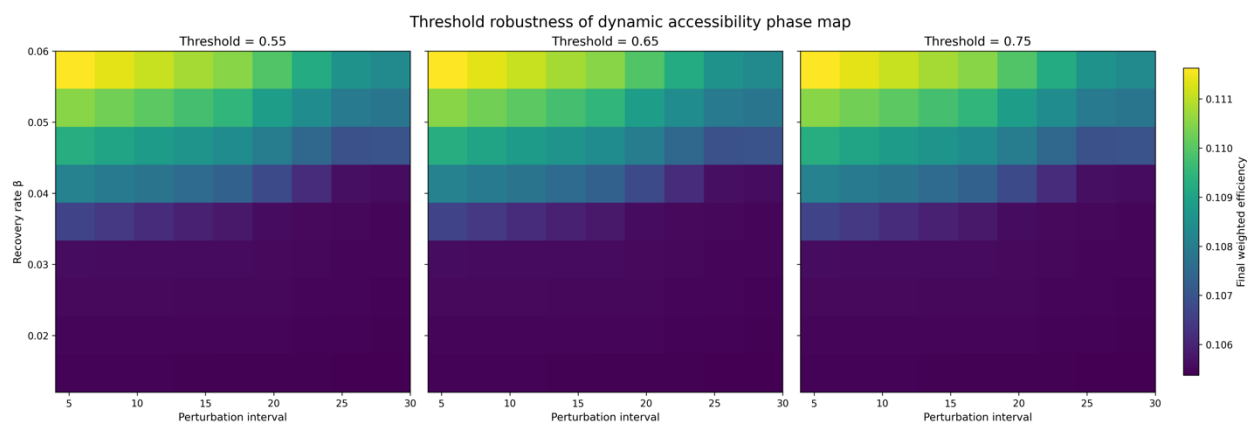


Figure 9. Across explored threshold choices, accessibility organization remains qualitatively observable despite moderate variation in regime boundaries.

Operational interpretation

Observed organization appears partially robust to threshold specification rather than emerging from a single threshold choice alone.

9.2 Null control: fixed recovery

Operational question

Does accessibility organization persist when recovery heterogeneity is removed?

To examine whether recovery dynamics contribute to accessibility organization, the notebook evaluates a simplified control in which recovery conditions are operationally homogenized.

This manipulation intentionally weakens recovery-dependent variability.

Figure 10 — Null control: fixed recovery

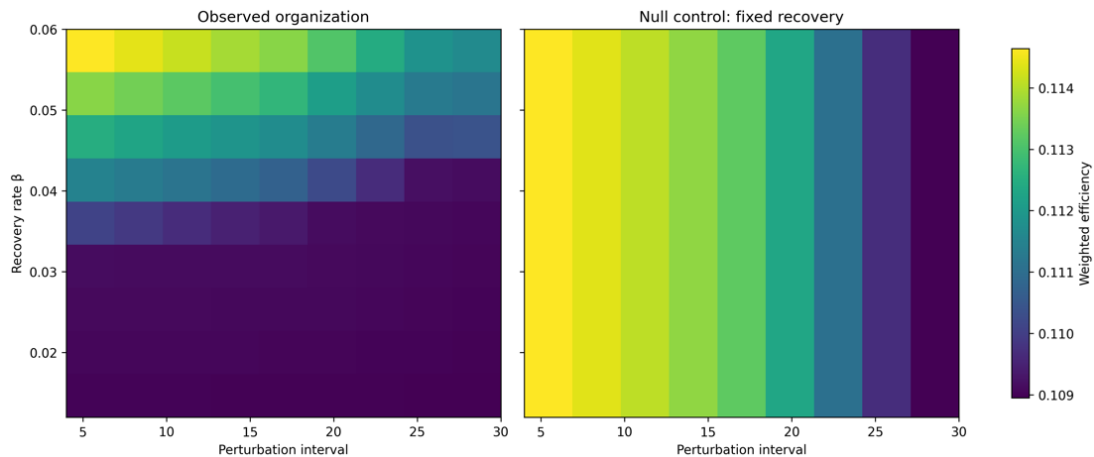


Figure 10. Under fixed recovery assumptions, accessibility organization becomes substantially weakened and progressively more timing-dominated.

Operational interpretation

Recovery kinetics may contribute operationally to shaping accessibility organization rather than acting as descriptive parameters alone.

9.3 Null control: reduced perturbation-history persistence

Operational question

Does accessibility organization persist when perturbation history is weakened?

To evaluate whether perturbation history contributes to accessibility organization, the notebook examines a control in which perturbation-history persistence is substantially reduced.

This manipulation weakens accumulated perturbation effects while preserving broader scaffold structure.

Figure 11 — Reduced perturbation-history persistence

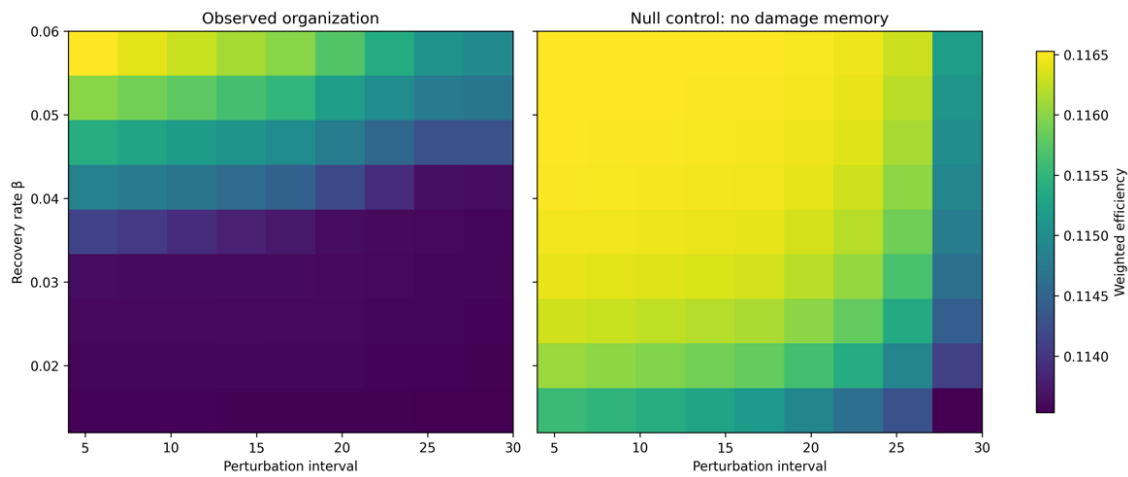


Figure 11. Accessibility organization remains partially observable under reduced perturbation-history persistence but becomes smoother and less sharply separated.

Operational interpretation

Perturbation history may sharpen accessibility organization rather than generate it de novo.

9.4 Stochastic robustness

Operational question

Does accessibility organization remain observable under stochastic perturbation variability?

To examine whether organization depends on highly deterministic conditions, moderate perturbation and recovery variability are introduced.

The objective is intentionally conservative: **to test whether qualitative organization survives stochastic perturbation.**

Figure 12 — Stochastic robustness

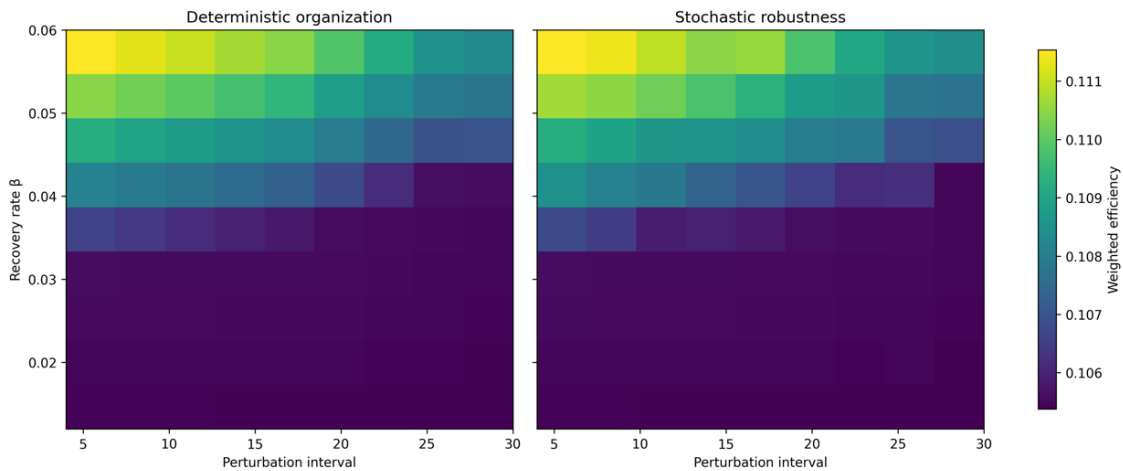


Figure 12. Across explored conditions, accessibility organization remains partially observable under stochastic perturbation variability, although boundaries become smoother and less sharply defined.

Operational interpretation

Observed organization may represent a dynamically admissible pattern rather than a purely deterministic artifact.

10. Operational regime organization

The previous sections examined whether accessibility organization remains observable across perturbation–recovery conditions and whether such organization survives robustness and falsification-oriented controls.

A natural operational question then becomes: **Can accessibility organization be summarized into interpretable operational regions?**

To facilitate interpretation, the notebook introduces a heuristic classification of accessibility organization across explored parameter space.

Importantly, this classification is **not intended to represent physical phases, hard boundaries, or universal system states**.

Instead, the objective is intentionally operational: **to summarize how accessibility reorganizes under perturbation–recovery constraints**.

Regions are heuristically classified using accessibility-related observable including:

- weighted accessibility efficiency,
- accessibility cost,

- reachable future fraction.

Figure 13 — Operational accessibility regime organization

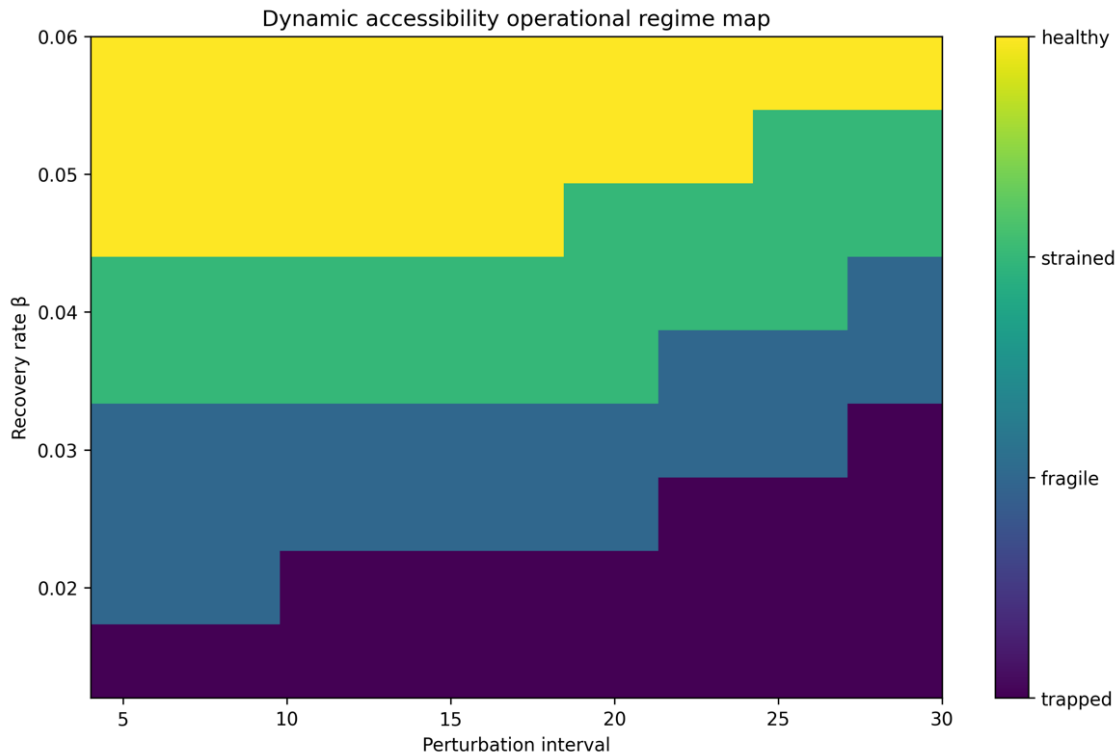


Figure 13. The regime map summarizes accessibility organization into interpretable operational regions under explored perturbation–recovery conditions.

The resulting map provides four visualization-oriented operational categories:

- **Healthy** — accessibility remains comparatively preserved, with lower coordination cost and higher weighted efficiency.
- **Strained** — accessibility remains operationally present but increasingly constrained.
- **Fragile** — accessibility becomes progressively more vulnerable to perturbation and recovery mismatch.
- **Trapped** — accessibility becomes substantially contracted under explored perturbation–recovery conditions.

These categories should be interpreted cautiously.

They are visualization-oriented operational abstractions rather than hard physical boundaries.

Under this framing, the regime map should be interpreted as an exploratory operational summary of accessibility organization within the assumptions of the notebook rather than as a predictive or domain-specific classification.

11. Limitations

The present notebook should be interpreted as an exploratory computational abstraction rather than a domain-specific explanatory model.

Its primary objective is operational: ***to test whether accessibility organization may emerge under perturbation–recovery constraints within a minimal scaffold-preserving framework***

Accordingly, several limitations should be stated explicitly.

11.1 Minimal operational abstraction

The implemented system represents a deliberately simplified computational scaffold.

The framework is designed to explore accessibility organization under perturbation–recovery constraints rather than reproduce the mechanistic complexity of biological, physical, ecological, or engineered systems.

The notebook should therefore be interpreted as a minimal exploratory abstraction rather than a realistic system model.

11.2 No empirical fitting

The explored dynamics are not calibrated against empirical measurements.

Parameters are selected to evaluate operational behavior under controlled perturbation–recovery conditions rather than reproduce observed experimental systems.

The present results should not be interpreted as fitted descriptions of real systems.

Instead, the notebook evaluates whether accessibility organization may emerge computationally under constrained assumptions.

11.3 Heuristic observables and thresholds

Several observables used throughout the notebook — including accessibility thresholds and regime classifications — are operational and heuristic.

In particular, regime categories are introduced to summarize organization visually rather than infer hard physical boundaries.

Accessibility regions should be interpreted as descriptive operational summaries rather than predictive categories.

11.4 Descriptive rather than predictive scope

The notebook is designed to characterize organization under explored perturbation–recovery conditions.

It does not attempt to predict system-specific outcomes, infer domain-specific mechanisms, or establish universal accessibility laws.

The framework evaluates whether accessibility organization appears computationally admissible under constrained assumptions.

11.5 No claim of universality

Although accessibility organization appears observable across explored perturbation–recovery conditions, no claim is made that such organization generalizes universally across domains.

Broader interpretation requires confrontation with empirical systems and domain-specific validation.

The notebook should therefore be interpreted as an exploratory operational hypothesis rather than a universal explanatory framework.

12. Reproducibility

This exploratory computational note accompanies a fully executable and reproducible notebook.

The associated repository includes:

- the executable notebook,
- generated figures,
- exported data tables,
- parameterized perturbation–recovery workflows,
- robustness and falsification-oriented controls,
- and operational regime summaries.

The notebook was designed to preserve execution transparency through:

- fixed computational order,
- explicit parameterization,
- exported intermediate observables,
- and reproducible analysis workflows.

Importantly: ***the notebook should be considered the primary computational object, while the present document serves as a methodological and interpretive companion.***

The purpose of the repository is to facilitate:

- direct inspection of assumptions,
- computational reproducibility,
- parameter modification,
- and falsification-oriented exploration of operational accessibility organization.

13. Conclusion

This exploratory notebook examined whether systems may preserve structural scaffold while progressively reorganizing operational accessibility under perturbation–recovery constraints.

Across explored conditions, accessibility-related observables — including reachable future fraction, accessibility cost, weighted efficiency, perturbation timing, recovery kinetics, and perturbation-history effects — appeared to display systematic operational variation while scaffold structure remained preserved.

Importantly, the notebook does not suggest that accessibility degradation necessarily requires explicit structural disconnection.

Instead, the explored simulations are consistent with a narrower operational interpretation:

preserved scaffold does not necessarily imply preserved accessibility.

Likewise, under explored perturbation–recovery conditions: **identical perturbation burden does not necessarily imply identical accessibility futures.**

Across the explored parameter ranges, accessibility organization appeared sensitive to the interaction between perturbation timing, recovery kinetics, and perturbation history.

At the same time, the framework should be interpreted cautiously.

The notebook represents a deliberately minimal computational abstraction rather than a fitted model of biological, physical, ecological, or engineered systems, and no claim is made regarding universal mechanisms or domain-specific causal laws.

Instead, the present work should be interpreted as a falsification-oriented computational exploration asking a narrower operational question: **may accessibility organization emerge operationally under perturbation–recovery constraints while scaffold remains preserved?**

A natural next step is confrontation with empirical dynamical systems to evaluate whether comparable accessibility organization becomes observable under experimentally measurable conditions.

Appendix A — Computational parameters and implementation details

The notebook implements a minimal scaffold-preserving perturbation–recovery framework designed to evaluate whether accessibility organization may emerge operationally under repeated perturbation.

The following parameters and observables are used throughout the notebook.

Parameter / observable	Operational role
Lattice size	Fixed interaction scaffold used throughout simulations
Perturbation interval	Temporal spacing between perturbation events
Recovery β	Recovery kinetics parameter controlling restoration dynamics
Accessibility threshold	Operational criterion defining effective traversability
Reachable future fraction	Fraction of scaffold remaining operationally reachable from a fixed source
Weighted efficiency	Summary observable of accessibility organization efficiency
Mean accessibility cost	Average operational burden required to traverse accessible regions
Worst-case accessibility cost	Maximum operational burden among accessible paths
Stochastic perturbation	Moderate perturbation variability used for robustness analysis
Perturbation history	Damage persistence / recovery-memory condition

Parameter sweeps, robustness analyses, null controls, and exported observables are fully implemented within the executable notebook accompanying this document.

Importantly, the notebook should be treated as the primary computational object, while the present appendix serves only as a reproducibility-oriented summary of operational assumptions and implementation structure.

References

1. Newman, M. (2010). *Networks: An Introduction*. Oxford University Press.
2. Barabási, A.-L. (2016). *Network Science*. Cambridge University Press.
3. Ashwin, P., Wieczorek, S., Vitolo, R., & Cox, P. (2012). *Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system*. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **370**(1962), 1166–1184.
4. Ritchie, P., & Sieber, J. (2016). *Early-warning indicators for rate-induced tipping*. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, **26**(9), 093116.
5. Wieczorek, S., Ashwin, P., Luke, C., & Cox, P. (2023). *Excitability, tipping, and transient dynamics in complex systems*. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **381**(2267), [20230018](#).
6. Rule, A., Birmingham, A., Zuniga, C., Altintas, I., Huang, S.-C., Knight, R., Moshiri, N., Nguyen, M. H., Rosenthal, S. B., Pérez, F., & Rose, P. W. (2019). *Ten simple rules for writing and sharing computational analyses in Jupyter Notebooks*. *PLOS Computational Biology*, **15**(7), e1007007.